

Legendre's Conjecture in Function Fields: Full Monodromy, Twisted Root Varieties, and Explicit Bounds

Revised version (v6) of *The Geometry of Prime Vacuums*

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<https://github.com/Ruqing1963/legendre-function-field>

Abstract

For a monic polynomial $f \in \mathbb{F}_q[t]$ of degree $d \geq 1$, the *Legendre interval* $\mathcal{I}_f = \{f^2 + s : \deg s \leq d\}$ is the natural function field analogue of $[n^2, (n+1)^2]$. That $\mathcal{I}_f$ contains $q^{d+1}/(2d) + O_d(q^{d+1/2})$ irreducible polynomials is a special case ($\epsilon = 1/2$) of the short-interval theorem of Bank, Bary-Soroker and Rosenzweig (2015); the earlier version (v5) of this preprint presented it as new, and this revision corrects that. We give a self-contained proof that the geometric monodromy group of the family is the full symmetric group S_{2d} whenever $p = \text{char } \mathbb{F}_q > 2d$, *without* the squarefree hypothesis on f that v5 required. We then prove an exact identity $2d N_{\text{irr}} = \#W(\mathbb{F}_q) - E$, where W is an explicit $(d+1)$ -dimensional twisted form of the root variety of the family, of degree at most $(d-1)!$, and $0 \leq E < q^{d+1}/(q-1)$. Combined with the explicit Lang–Weil estimate of Cafure and Matera, this yields a fully explicit and rigorous threshold $q_0(d)$ beyond which every Legendre interval contains an irreducible polynomial. For $d \leq 3$ we obtain closed formulas valid for every admissible q ; for $d = 3$ they agree with Kuz'min's count of irreducible polynomials with two prescribed coefficients. Exhaustive and sampled computations for $d \leq 6$ support every statement, and the code and data are public. An appendix lists the errors in v5 and how each one is fixed here.

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1 Introduction

1.1 Background

Legendre’s conjecture asserts that for every integer $n \geq 1$ there is a prime between n^2 and $(n + 1)^2$. It is open; the best unconditional result on prime gaps (Baker–Harman–Pintz [1]) gives primes in $[x, x + x^{0.525}]$, whereas the Legendre interval has length about $2x^{1/2}$.

In $\mathbb{F}_q[t]$ the absolute value is $|g| = q^{\deg g}$. For monic f of degree d the polynomials g with $|g - f^2| \leq |f|$ are exactly

$$\mathcal{I}_f = \{ f^2 + s : s \in \mathbb{F}_q[t], \deg s \leq d \}, \quad |\mathcal{I}_f| = q^{d+1},$$

an interval of “length” $|f| = |f^2|^{1/2}$ around f^2 , the analogue of $[n^2, n^2 + n] \subset [n^2, (n + 1)^2]$. All elements of $\mathcal{I}_f$ are monic of degree $2d$. The prime polynomial theorem predicts about $|\mathcal{I}_f|/(2d)$ irreducibles.

Definition 1.1. Write $\mathcal{M} = \mathbb{A}^{d+1}$ with coordinates $\mathbf{a} = (a_0, \dots, a_d)$ and put

$$F(t, \mathbf{a}) = f(t)^2 + \sum_{i=0}^d a_i t^i, \quad N_{\text{irr}}(f) = \#\{\mathbf{a} \in \mathbb{F}_q^{d+1} : F(t, \mathbf{a}) \text{ irreducible in } \mathbb{F}_q[t]\}.$$

1.2 What was already known

Bank, Bary-Soroker and Rosenzweig [2] proved that for every partition λ of k , every monic f of degree k and every m with $1 \leq m < k$ (with $m \geq 2$ if $p \mid k(k - 1)$, and $m \geq 3$ if $p = 2$ and $\deg f' \leq 1$), the number of $g \in f + \mathbb{F}_q[t]_{\leq m}$ with factorization type λ is $P(\lambda)q^{m+1} + O_k(q^{m+1/2})$, where $P(\lambda)$ is the proportion of permutations of cycle type λ in S_k . Taking $k = 2d$, $m = d$ and replacing f by f^2 gives, for every odd q ,

$$N_{\text{irr}}(f) = \frac{q^{d+1}}{2d} + O_d(q^{d+1/2}). \tag{1}$$

This is exactly the “Main Theorem” of v5 of this preprint, which therefore is not new. The implied constant in [2] is not explicit.

1.3 Results of this paper

Throughout, q is a power of a prime p , and $f \in \mathbb{F}_q[t]$ is monic of degree $d \geq 1$. We write G_{geom} for the Galois group of $F(t, \mathbf{a})$ over $\overline{\mathbb{F}}_q(\mathbf{a})$, i.e. the geometric monodromy group of the finite cover $\{F = 0\} \rightarrow \mathcal{M}$ (Section 3).

Theorem A (Section 3). *If $p > 2d$, then $G_{\text{geom}} = S_{2d}$. No assumption on f beyond monicity is needed.*

Theorem B (Section 4). *There is an affine variety W over $\mathbb{F}_q$, explicitly defined by $d - 1$ equations of degrees $1, 2, \dots, d - 1$ in $\mathbb{A}^{2d}$, such that for all q*

$$2d \cdot N_{\text{irr}}(f) = \#W(\mathbb{F}_q) - E(f), \quad 0 \leq E(f) \leq \sum_{e|2d, e < 2d} q^e < \frac{q^{d+1}}{q-1}.$$

Its $\mathbb{F}_q$ -points are the $\beta \in \mathbb{F}_{q^{2d}}$ whose characteristic polynomial lies in $\mathcal{I}_f$. W has pure dimension $d + 1$ and degree at most $(d - 1)!$, and it is absolutely irreducible if and only if $G_{\text{geom}} = S_{2d}$.

Theorem C (Section 5). *Assume $G_{\text{geom}} = S_{2d}$ (for instance $p > 2d$). Let $\delta = (d - 1)!$. If $q > 2(d + 2)\delta^2$, then*

$$|2d N_{\text{irr}}(f) - q^{d+1}| < (\delta - 1)(\delta - 2) q^{d+1/2} + (5\delta^{13/3} + 2) q^d.$$

Consequently $N_{\text{irr}}(f) \geq 1$ for all $q > q_0(d) := \max\{2(d + 2)\delta^2, ((\delta - 1)(\delta - 2) + \sqrt{5\delta^{13/3} + 2})^2\}$.

Theorem D (Section 6). *(i) If $d = 1$ then $N_{\text{irr}} = (q^2 - q)/2$. (ii) If $d = 2$ and q is odd then $N_{\text{irr}} = (q^3 - q)/4$. (iii) If $d = 3$ and $p \geq 5$, write $f = t^3 + c_2 t^2 + c_1 t + c_0$ and $\Delta = c_2^2 - 3c_1$; let η be the quadratic character of $\mathbb{F}_q$. Then*

$$6 N_{\text{irr}} = \begin{cases} q^4 - \eta(2\Delta) q^2 - q + \eta(-3) + \eta(2\Delta) - 1, & \Delta \neq 0, \\ q^4 - q - (q - 1) \eta(-3), & \Delta = 0. \end{cases}$$

In particular every Legendre interval with $d \leq 3$ contains an irreducible polynomial, for every q in the stated range.

Theorem C gives, for example, $q_0(4) \approx 1.65 \times 10^4$ and $q_0(5) \approx 7.3 \times 10^6$ (Table 1). For $d \leq 3$ no threshold is needed. Theorem D(i)–(ii) are classical consequences of the counts of irreducible polynomials with prescribed trace [5], and (iii) is a special case of Kuz'min's formula for two prescribed coefficients (see [13, 15]). We include short proofs because they come out of Theorem B directly and they make good test cases for the computations.

The genuinely hard analogue of Legendre's conjecture fixes q and lets $d \rightarrow \infty$. Then one must prescribe the top $d - 1$ coefficients of a polynomial of degree $2d$, about half of them. Existing results on irreducible polynomials with prescribed coefficients [17, 10] handle far fewer coefficients, so this regime remains open (Section 8).

1.4 About this revision

Version v5 [6] contained several gaps and incorrect citations, listed in Appendix A. The most serious were: the main theorem was not new; the primitivity argument used a criterion that does not apply as stated; the effective Chebotarev bound was formulated with the wrong sheaf; and the explicit threshold $(8d)^{2d+6}$ relied on a Betti-number estimate that is not in the cited source. The present version replaces the cohomological error analysis by the elementary twisted-variety identity (Theorem B) and an explicit Lang–Weil bound, so that every constant is proved.

2 Preliminaries

For $\beta \in \mathbb{F}_{q^n}$ let $\chi_\beta(t) = \prod_{k=0}^{n-1} (t - \beta^{q^k}) \in \mathbb{F}_q[t]$ be its characteristic polynomial over $\mathbb{F}_q$. If β has degree $e \mid n$ over $\mathbb{F}_q$ and minimal polynomial μ_β , then $\chi_\beta = \mu_\beta^{n/e}$.

Lemma 2.1 (Membership in $\mathcal{I}_f$). *A monic $P \in \mathbb{F}_q[t]$ of degree $2d$ lies in $\mathcal{I}_f$ if and only if its coefficients of $t^{2d-1}, \dots, t^{d+1}$ agree with those of f^2 . If $p > d - 1$, this holds if and only if the power sums $p_j(P) = \sum_{P(\rho)=0} \rho^j$ satisfy $p_j(P) = 2p_j(f)$ for $1 \leq j \leq d - 1$.*

Proof. The first statement is the definition. For the second, Newton's identities express the power sums $p_1, \dots, p_{d-1}$ through the top $d - 1$ coefficients. The converse direction needs division by $1, \dots, d - 1$, which is allowed when $p > d - 1$. Finally $p_j(f^2) = 2p_j(f)$. $\square$

Lemma 2.2 (Inertia at a simple node). *Let R be a complete discrete valuation ring with algebraically closed residue field k of characteristic $\neq 2$, fraction field K and valuation v . Let $P \in R[t]$ be monic of degree n whose reduction $\bar{P}$ has exactly one double root β and $n - 2$ simple roots, and suppose $v(\text{disc} P) = 1$. Then the splitting field of P over K is a quadratic extension whose Galois group acts on the roots of P as a transposition.*

Proof. Write $\bar{P} = (t - \beta)^2 \bar{L}$ with $\bar{L}(\beta) \neq 0$ and $\bar{L}$ squarefree. By Hensel's lemma $P = QL$ with $Q, L \in R[t]$ monic lifting $(t - \beta)^2$ and $\bar{L}$, and L splits into linear factors over R . Since $\text{disc} P = \text{disc} Q \cdot \text{disc} L \cdot \text{Res}(Q, L)^2$ and both $\text{disc} L$ and $\text{Res}(Q, L)$ are units, $v(\text{disc} Q) = 1$. An element of odd valuation is not a square in K , so Q is irreducible over K and its splitting field is a quadratic extension. Its nontrivial automorphism swaps the two roots of Q and fixes the roots of L . $\square$

Lemma 2.3 (Primitivity and decomposition; cf. [18, 16]). *Let K be a field, $G \in K[t]$ of degree $n \geq 2$ with $G' \neq 0$, and x transcendental over K . If $\text{Gal}(G(t) - x / K(x))$, acting on the roots of $G(t) - x$, is imprimitive, then $G = g \circ h$ with $g, h \in K[t]$ and $\deg g, \deg h \geq 2$.*

Proof. Let τ be a root and Γ the Galois group. Blocks containing τ correspond to subgroups between the stabilizer of τ and Γ , hence to fields $K(x) \subseteq L \subseteq K(\tau)$. Imprimitivity gives a proper intermediate field L , and $L = K(h)$ for some $h \in K(\tau)$ by Lüroth's theorem. Since $x = G(\tau)$ is a polynomial in τ , the place $\tau = \infty$ is the only place of $K(\tau)$ above $x = \infty$; it is K -rational. Its restriction to L is therefore the only place of L above $x = \infty$, and it is K -rational. After a Möbius transformation over K we may take it to be $h = \infty$. Then h , as a function of τ , has its only pole at $\tau = \infty$, so $h \in K[\tau]$. Likewise $x \in K(h)$ has its only pole at $h = \infty$, so $x = g(h)$ with $g \in K[X]$. Both degrees are ≥ 2 because L is a proper intermediate field. $\square$

3 The monodromy group

Let $\mathcal{H} = \{F(t, \mathbf{a}) = 0\} \subset \mathcal{M} \times \mathbb{A}^1$ and $\pi : \mathcal{H} \rightarrow \mathcal{M}$ the projection, a finite flat morphism of degree $2d$. Let $\mathcal{D} \subset \mathcal{M}$ be the zero locus of $\text{disc}_t F$ and $\mathcal{M}^\circ = \mathcal{M} \setminus \mathcal{D}$. Over $\mathcal{M}^\circ$, π is finite étale. Its geometric monodromy group is the image of $\pi_1(\mathcal{M}_{\mathbb{F}_q}^\circ)$ in S_{2d} , which equals $G_{\text{geom}} = \text{Gal}(F / \mathbb{F}_q(\mathbf{a}))$. Put $K' = \mathbb{F}_q(a_1, \dots, a_d)$ and

$$G(t) = f(t)^2 + a_1 t + \dots + a_d t^d \in K'[t], \quad \text{so} \quad F = G(t) + a_0.$$

Since a_0 is transcendental over K' , $G_{\text{geom}} = \text{Gal}(G(t) - x / K'(x))$ with $x = -a_0$.

Proposition 3.1 (Transitivity). *F is irreducible over $\mathbb{F}_q(\mathbf{a})$, so G_{geom} is transitive.*

Proof. F has degree one in a_0 with coefficients 1 and $f^2 + \sum_{i \geq 1} a_i t^i$, which are coprime. Hence F is irreducible in $\overline{\mathbb{F}}_q[\mathbf{a}, t]$, and by Gauss's lemma it is irreducible in $\overline{\mathbb{F}}_q(\mathbf{a})[t]$ because it is monic in t . $\square$

Proposition 3.2 (Primitivity). *If $p > d$, then G_{geom} is primitive.*

Proof. For $d = 1$ this is automatic since $2d = 2$. Let $d \geq 2$. Note $G' \neq 0$ because of the term $a_1 t$. Suppose G_{geom} is imprimitive. By Lemma 2.3 applied over K' , $G = g \circ h$ with $g, h \in K'[t]$, $\deg h = m$, $\deg g = n$, $mn = 2d$ and $m, n \geq 2$. In particular $m \leq d$. Replacing h by $\lambda h + \mu$ and g accordingly, we may assume h monic with $h(0) = 0$. Write $h = t^m + h_{m-1}t^{m-1} + \dots + h_1 t$.

Step 1: h has constant coefficients. For $1 \leq j \leq m-1$ we have $2d-j \geq 2d-m+1 \geq d+1$, so the coefficient of t^{2d-j} in G is the corresponding coefficient of f^2 , an element of $\mathbb{F}_q$. On the other side, $g(h) = c h^n + (\text{terms of degree} \leq 2d-m)$ with c the leading coefficient of g , and $c = 1$ because G is monic. Expanding h^n gives

$$[t^{2d-j}] h^n = n h_{m-j} + \Phi_j(h_{m-1}, \dots, h_{m-j+1}),$$

with Φ_j a polynomial with integer coefficients. Since $2 \leq n \leq d < p$, n is invertible in $\mathbb{F}_q$. Induction on j shows $h_{m-1}, \dots, h_1 \in \mathbb{F}_q$, so $h \in \mathbb{F}_q[t]$.

Step 2: contradiction. Let $U = \text{span}_{\mathbb{F}_q}\{1, h, \dots, h^n\} \subset \mathbb{F}_q[t]_{\leq 2d}$. Since $G = \sum_k g_k h^k$ with $g_k \in K'$, we get $G \in U \otimes_{\mathbb{F}_q} K'$. Let ℓ be any $\mathbb{F}_q$ -linear form on $\mathbb{F}_q[t]_{\leq 2d}$ that vanishes on U , extended K' -linearly. Then $0 = \ell(G) = \ell(f^2) + \sum_{i=1}^d a_i \ell(t^i)$ in K' , and since $a_1, \dots, a_d$ are algebraically independent, $\ell(t) = 0$. As this holds for every such ℓ , $t \in U$. But every nonconstant element of U has degree divisible by $m \geq 2$. Contradiction. $\square$

Proposition 3.3 (A transposition). *If $p > 2d$, then G_{geom} contains a transposition.*

Proof. Step 1: a fibre with exactly one double root. If $d = 1$, take $P^* = (t - \beta)^2$ for any $\beta \in \overline{\mathbb{F}}_q$; it lies in $\mathcal{I}_f \otimes \overline{\mathbb{F}}_q$ since $\deg((t - \beta)^2 - f^2) \leq 1$. Let $d \geq 2$ and fix $\beta \in \overline{\mathbb{F}}_q$. Divide $f^2 = (t - \beta)^2 Q + r$ with $\deg r \leq 1$; Q is monic of degree $2d - 2$. For $c \in \overline{\mathbb{F}}_q$ put $R_c = Q + c$ and $P^* = (t - \beta)^2 R_c$. Then $P^* - f^2 = c(t - \beta)^2 - r$ has degree $\leq 2 \leq d$, so P^* is a fibre of F over some $\mathbf{a}^* \in \mathcal{M}(\overline{\mathbb{F}}_q)$. The derivative Q' has leading coefficient $2d - 2 \not\equiv 0 \pmod{p}$, so R_c has a repeated root only if $c = -Q(\gamma)$ for a root γ of Q' . Excluding these finitely many values and $c = -Q(\beta)$, the polynomial R_c is squarefree and $R_c(\beta) \neq 0$. So $P^* = (t - \beta)^2 R$ with R squarefree and $R(\beta) \neq 0$ (with $R = 1$ if $d = 1$).

Step 2: the discriminant vanishes to order one. Consider the line $\varepsilon \mapsto \mathbf{a}^* + \varepsilon e_0$, along which $F = P^* + \varepsilon$. Since $\partial_t F = P^{*'}$ does not depend on ε and has leading coefficient $2d \not\equiv 0$ in $\mathbb{F}_q$, we have

$$\text{disc}_t(P^* + \varepsilon) = u \prod_{\gamma: P^{*'}(\gamma)=0} (P^*(\gamma) + \varepsilon),$$

with $u \in \overline{\mathbb{F}}_q^\times$ and roots γ counted with multiplicity. Its order of vanishing at $\varepsilon = 0$ is the number of common roots of P^* and $P^{*'}$, counted with their multiplicity in $P^{*'}$. Now $P^{*' } = (t - \beta)(2R + (t - \beta)R')$. The root β is simple in $P^{*'}$ because $2R(\beta) \neq 0$. A root ρ of R is not a root of $P^{*'}$, because $P^{*' }(\rho) = (\rho - \beta)^2 R'(\rho) \neq 0$. Hence the order is exactly 1. In particular the line is not contained in $\mathcal{D}$.

Step 3: inertia. The Galois group of F restricted to the line is a subgroup of G_{geom} . It contains the inertia group at $\varepsilon = 0$, which is the Galois group of $P^* + \varepsilon$ over $\overline{\mathbb{F}}_q((\varepsilon))$. By Lemma 2.2 it is generated by a transposition. $\square$

Theorem 3.4 (Theorem A). *If $p > 2d$, then $G_{\text{geom}} = S_{2d}$, and also $\text{Gal}(F/\mathbb{F}_q(\mathbf{a})) = S_{2d}$.*

Proof. G_{geom} is primitive (Propositions 3.1 and 3.2) and contains a transposition (Proposition 3.3). A primitive permutation group containing a transposition is the full symmetric group [8, Thm. 3.3A]. The arithmetic group contains G_{geom} and lies in S_{2d} . $\square$

Remark 3.5. (1) No squarefree hypothesis on f is used. In v5 it served only to locate a simple branch point near $\mathbf{a} = 0$, and that argument had gaps (Appendix A). (2) The hypothesis $p > 2d$ is used to invert $n \leq d$, to make $2d$ and $2d - 2$ nonzero in $\mathbb{F}_q$, and to have $p \neq 2$. The result $G_{\text{geom}} = S_{2d}$ also follows from [2, Prop. 3.6] for every odd p .

4 The twisted root variety

4.1 The root variety

Let $e_j(u)$ denote the j -th elementary symmetric polynomial in $u = (u_1, \dots, u_{2d})$, and $\gamma_j = (-1)^j [t^{2d-j}] f^2 \in \mathbb{F}_q$. Define

$$V = \{u \in \mathbb{A}^{2d} : e_j(u) = \gamma_j, 1 \leq j \leq d-1\}.$$

Let $\Phi : \mathbb{A}_u^{2d} \rightarrow \mathbb{A}^{2d}$ send u to the coefficients of $\prod_i (t - u_i)$. Φ is finite, flat and surjective, and S_{2d} permutes the coordinates of its fibres. Identify $\mathcal{M}$ with the affine subspace $\mathcal{I}_f$ of monic polynomials of degree $2d$. Then $V = \Phi^{-1}(\mathcal{M})$.

Proposition 4.1. (a) V is a complete intersection of pure dimension $d+1$ and degree at most $(d-1)!$.

(b) V is reduced, and $V^\circ := V \setminus \Phi^{-1}(\mathcal{D})$ is smooth and dense in V .

(c) $V_{\overline{\mathbb{F}}_q}$ is irreducible if and only if $G_{\text{geom}} = S_{2d}$.

Proof. (a) Every component of $V_{\overline{\mathbb{F}}_q}$ has dimension at least $2d - (d-1) = d+1$ by Krull's theorem, and at most $\dim \mathcal{M} = d+1$ because Φ is finite. So V is a complete intersection and hence Cohen–Macaulay. The refined Bézout inequality [9, Example 8.4.6] gives $\deg V \leq \prod_{j < d} j = (d-1)!$.

(b) The Jacobian of Φ is the Vandermonde determinant, so Φ is étale where the u_i are distinct. Hence $V^\circ \rightarrow \mathcal{M}^\circ$ is étale and V° is smooth. $\Phi^{-1}(\mathcal{D}) \cap V$ has dimension $\leq d$, so it contains no component of V . Thus V is generically reduced and Cohen–Macaulay, hence reduced.

(c) Over $\mathcal{M}^\circ$, V° is an S_{2d} -torsor: S_{2d} acts simply transitively on each fibre, the orderings of the $2d$ distinct roots. Choosing one ordering of the base fibre identifies it with S_{2d} , and $\pi_1(\mathcal{M}_{\overline{\mathbb{F}}_q}^\circ)$ acts by left multiplication through its image G_{geom} . Hence $V_{\overline{\mathbb{F}}_q}^\circ$ has $[S_{2d} : G_{\text{geom}}]$ connected components. Since V° is smooth and dense, $V_{\overline{\mathbb{F}}_q}$ is irreducible if and only if this index is 1. $\square$

4.2 The twist

Fix an $\mathbb{F}_q$ -basis $\omega_1, \dots, \omega_{2d}$ of $\mathbb{F}_{q^{2d}}$ and let $M = (\omega_i^{q^{k-1}})_{k,i}$ be the Moore matrix, which is invertible. Define $W \subset \mathbb{A}_x^{2d}$ by the equations $e_j(Mx) = \gamma_j$, $1 \leq j \leq d-1$.

Lemma 4.2. W is defined over $\mathbb{F}_q$, and $x \mapsto Mx$ is an isomorphism $W_{\mathbb{F}_{q^{2d}}} \cong V_{\mathbb{F}_{q^{2d}}}$. The map $x \mapsto \beta = \sum_i x_i \omega_i$ is a bijection

$$W(\mathbb{F}_q) \xrightarrow{\sim} \{\beta \in \mathbb{F}_{q^{2d}} : \chi_\beta \in \mathcal{I}_f\}.$$

Proof. Put $u_k = \sum_i \omega_i^{q^{k-1}} x_i$. Raising the coefficients of u_k to the q -th power gives u_{k+1} , indices mod $2d$. So Frobenius on coefficients permutes $u_1, \dots, u_{2d}$ cyclically and fixes each $e_j(u)$. Hence

$e_j(Mx) \in \mathbb{F}_q[x]$. The second claim is clear because M is invertible over $\mathbb{F}_{q^{2d}}$. For $x \in \mathbb{F}_q^{2d}$ we have $u = (\beta, \beta^q, \dots, \beta^{q^{2d-1}})$, and $e_j(u)$ is $(-1)^j$ times the coefficient of t^{2d-j} in χ_β . Apply Lemma 2.1. $\square$

Theorem 4.3 (Theorem B). *For every monic f of degree $d \geq 1$ and every q ,*

$$2d \cdot N_{\text{irr}}(f) = \#W(\mathbb{F}_q) - E(f), \quad E(f) = \#\left\{\beta \in \bigcup_{e|2d, e < 2d} \mathbb{F}_{q^e} : \chi_\beta \in \mathcal{I}_f\right\}.$$

Moreover $0 \leq E(f) \leq \sum_{e|2d, e < 2d} q^e < q^{d+1}/(q-1)$. W is an affine $\mathbb{F}_q$ -variety of dimension $d+1$ and degree at most $(d-1)!$, and it is absolutely irreducible if and only if $G_{\text{geom}} = S_{2d}$.

Proof. If β generates $\mathbb{F}_{q^{2d}}$, then $\chi_\beta = \mu_\beta$ is irreducible. Conversely, an irreducible $P \in \mathcal{I}_f$ has exactly $2d$ roots in $\mathbb{F}_{q^{2d}}$, and $\chi_\beta = P$ for each of them. Together with Lemma 4.2, this gives the identity. The bound on E comes from $\#\mathbb{F}_{q^e} = q^e$ and the fact that proper divisors of $2d$ are at most d . The remaining statements follow from Proposition 4.1 and Lemma 4.2, because a linear change of coordinates preserves dimension, degree and absolute irreducibility. $\square$

Remark 4.4. Theorem B is a twisted form of the Chebotarev argument. It counts the Frobenius class of a $2d$ -cycle by counting rational points on one twist of the S_{2d} -torsor V° . No étale cohomology of local systems is needed, and the error term reduces to a point count on an explicit, absolutely irreducible complete intersection.

5 An explicit bound

Theorem 5.1 (Cafure–Matera [4, Thm. 7.1]). *Let $V \subset \mathbb{A}^n$ be an absolutely irreducible $\mathbb{F}_q$ -variety of dimension $r > 0$ and degree δ . If $q > 2(r+1)\delta^2$, then $|\#V(\mathbb{F}_q) - q^r| < (\delta-1)(\delta-2)q^{r-1/2} + 5\delta^{13/3}q^{r-1}$.*

Theorem 5.2 (Theorem C). *Assume $G_{\text{geom}} = S_{2d}$, for example $p > 2d$. Let $\delta = (d-1)!$. If $q > 2(d+2)\delta^2$, then*

$$|2d N_{\text{irr}}(f) - q^{d+1}| < (\delta-1)(\delta-2)q^{d+1/2} + (5\delta^{13/3} + 2)q^d.$$

Proof. By Theorem B, W is absolutely irreducible of dimension $d+1$ and degree $\delta_W \leq \delta$. The bound in Theorem 5.1 and its hypothesis are monotone in the degree, so we may use δ in place of δ_W . Then $|2d N_{\text{irr}} - q^{d+1}| \leq |\#W(\mathbb{F}_q) - q^{d+1}| + E(f)$, and $E(f) < q^{d+1}/(q-1) \leq 2q^d$. $\square$

Corollary 5.3. *Under the hypotheses of Theorem 5.2, $N_{\text{irr}}(f) \geq 1$ for every*

$$q > q_0(d) = \max\left\{2(d+2)\delta^2, ((\delta-1)(\delta-2) + \sqrt{5\delta^{13/3} + 2})^2\right\}.$$

Proof. Put $A = (\delta-1)(\delta-2)$, $B = 5\delta^{13/3} + 2$ and $s = \sqrt{q}$. We need $s^2 - As - B > 0$. The left side increases for $s > A/2$, and at $s = A + \sqrt{B}$ it equals $A\sqrt{B} \geq 0$. $\square$

A second explicit route uses étale cohomology directly, with Katz's explicit bound on sums of Betti numbers.

Proposition 5.4. *Assume $G_{\text{geom}} = S_{2d}$ and put $\sigma_d = 3(d+1)^{3d-1}$. Then for every q ,*

$$|2d N_{\text{irr}}(f) - q^{d+1}| \leq (\sigma_d - 1)q^{d+1/2} + 2q^d,$$

and hence $N_{\text{irr}}(f) \geq 1$ for all $q > \sigma_d^2 = 9(d+1)^{6d-2}$.

d	v5 claim (unproved)	Katz + Deligne	this paper (Cor. 5.3)
1	7.2	2.2	0.8*
2	12.0	5.7	0.9*
3	16.6	10.6	2.0*
4	21.1	16.3	4.2
5	25.6	22.7	6.9
6	30.3	29.7	9.9
8	39.7	44.8	16.8
10	49.5	61.4	24.8
15	74.9	106.9	48.1
20	101.4	157.0	74.7
30	157.1	266.4	134.8
40	215.4	384.8	201.4
50	275.8	509.8	272.8

* not needed: Theorem D gives $N_{\text{irr}} \geq 1$ for all admissible q .

Table 1: Thresholds for $N_{\text{irr}} \geq 1$, as $\log_{10} q_0(d)$. The column “v5” is the value $(8d)^{2d+6}$ claimed in v5; it was not proved. The column “this paper” is Corollary 5.3. For $d \leq 3$ Theorem D removes the threshold. Source: `data/thresholds.csv`.

Proof. $W \subset \mathbb{A}^{2d}$ is defined by $d - 1$ equations of degree $\leq d - 1$. By [12, Theorem 12 and the remark following it], applied with no characters, the sum of the compactly supported Betti numbers of $W_{\overline{\mathbb{F}}_q}$ is at most $3(2 + (d - 1))^{2d+(d-1)} = \sigma_d$. By the Grothendieck–Lefschetz trace formula, $\#W(\mathbb{F}_q) = \sum_i (-1)^i \text{Tr}(\text{Frob} \mid H_c^i(W_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell))$. Since W is absolutely irreducible of dimension $d + 1$, H_c^{2d+2} is one-dimensional with eigenvalue q^{d+1} . Every other H_c^i is mixed of weights $\leq i \leq 2d + 1$ by Deligne [7, Thm. 3.3.1], so its eigenvalues have absolute value $\leq q^{d+1/2}$. This gives $|\#W(\mathbb{F}_q) - q^{d+1}| \leq (\sigma_d - 1)q^{d+1/2}$, and we finish with $E(f) \leq 2q^d$. For $s = \sqrt{q} > \sigma_d$ we have $s^2 - (\sigma_d - 1)s - 2 > \sigma_d - 2 > 0$. $\square$

Remark 5.5. Proposition 5.4 needs no lower bound on q for the estimate itself, but its threshold is larger than that of Corollary 5.3 for every d (Table 1). Both rigorous thresholds have the shape $d^{O(d)}$, like the unproved v5 value $(8d)^{2d+6}$. Corollary 5.3 is smaller than the v5 value for every $d \leq 52$; since $\log q_0(d) \sim \frac{13}{3} \log(d - 1)!$ while $\log(8d)^{2d+6} \sim 2d \log d$, the order reverses for large d . A sharper Betti-number bound for W would improve both.

6 Exact formulas for $d \leq 3$

Theorem 6.1 (Theorem D (i), (ii)). *If $d = 1$ then $N_{\text{irr}} = (q^2 - q)/2$. If $d = 2$ and q is odd then $N_{\text{irr}} = (q^3 - q)/4$, for every f .*

Proof. For $d = 1$, $W = \mathbb{A}^2$ and $E = \#\mathbb{F}_q = q$. For $d = 2$, W is the affine hyperplane $\text{Tr}_{\mathbb{F}_{q^4}/\mathbb{F}_q}(\beta) = \gamma_1$, which is a nonzero linear condition over $\mathbb{F}_q$, so $\#W(\mathbb{F}_q) = q^3$. If $\beta \in \mathbb{F}_{q^2}$, then $\chi_\beta = (\chi_\beta^{(2)})^2$ and the condition becomes $2\text{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(\beta) = \gamma_1$. This has exactly q solutions, since q is odd and the trace is surjective. Hence $E = q$. $\square$

Theorem 6.2 (Theorem D (iii)). *Let $d = 3$, $p \geq 5$, $f = t^3 + c_2 t^2 + c_1 t + c_0$ and $\Delta = c_2^2 - 3c_1$. Then $6N_{\text{irr}}$ is given by the formula in Theorem D. In particular $|6N_{\text{irr}} - q^4| \leq q^2 + q + 2$ and $N_{\text{irr}} \geq 1$.*

Proof. We use Lemma 2.1 with $s_1 = p_1(f) = -c_2$ and $s_2 = p_2(f) = c_2^2 - 2c_1$. For a finite extension $L/\mathbb{F}_q$ of degree n with $p \nmid n$, the trace form $B_L(x, y) = \text{Tr}_{L/\mathbb{F}_q}(xy)$ is nondegenerate. Its Gram determinant is $\det(M)^2$ for the Moore matrix M . Frobenius permutes the rows of M by an n -cycle, so $\det M \in \mathbb{F}_q$ if and only if n is odd. Hence $\eta(\det B_L) = (-1)^{n-1}$. The restriction of B_L to $H_L = \ker \text{Tr}_{L/\mathbb{F}_q}$ is nondegenerate with $\det(B_L|_{H_L}) \equiv n \det B_L$ modulo squares, because $L = \mathbb{F}_q \cdot 1 \perp H_L$ and $B_L(1, 1) = n$. We use the standard counts for a nondegenerate quadratic form Q in k variables over $\mathbb{F}_q$ [14, Thms. 6.26, 6.27]: $\#\{Q = b\} = q^{k-1} + q^{(k-1)/2}\eta((-1)^{(k-1)/2}b \det Q)$ if k is odd, and $\#\{Q = b\} = q^{k-1} + v(b)q^{(k-2)/2}\eta((-1)^{k/2} \det Q)$ if k is even, where $v(0) = q - 1$ and $v(b) = -1$ for $b \neq 0$.

Points of W . Here $L = \mathbb{F}_{q^6}$ and the conditions are $\text{Tr}\beta = 2s_1$ and $\text{Tr}\beta^2 = 2s_2$. Write $\beta = s_1/3 + \gamma$ with $\gamma \in H_L$. The second condition becomes $\text{Tr}(\gamma^2) = 2s_2 - \frac{2}{3}s_1^2 = \frac{4}{3}\Delta$. With $k = 5$ and $\eta(\det) = -\eta(6)$ we get $\#W(\mathbb{F}_q) = q^4 - \eta(8\Delta)q^2 = q^4 - \eta(2\Delta)q^2$.

The correction E . For $\beta \in \mathbb{F}_{q^3}$ the conditions read $\text{Tr}_3\beta = s_1$ and $\text{Tr}_3\beta^2 = s_2$. As above this is $\text{Tr}_3\gamma^2 = \frac{2}{3}\Delta$ on a 2-dimensional space with $\eta(\det) = \eta(3)$. This gives $E_3 = q - \eta(-3)$ if $\Delta \neq 0$ and $E_3 = q + (q-1)\eta(-3)$ if $\Delta = 0$. For $\beta \in \mathbb{F}_{q^2}$, write $\beta = s_1/3 + y\omega$ with $\omega^q = -\omega$, so $\omega^2 = n_0$ is a nonsquare in $\mathbb{F}_q$. The conditions reduce to $2n_0y^2 = \frac{4}{9}\Delta$. This gives $E_2 = 1 - \eta(2\Delta)$ if $\Delta \neq 0$ and $E_2 = 1$ otherwise. For $\beta \in \mathbb{F}_q$ the conditions force $\beta = s_1/3$ and $\Delta = 0$, so $E_1 = [\Delta = 0]$. Finally $E = E_3 + E_2 - E_1$, and Theorem B gives the formula. $\square$

Remark 6.3. For $d = 3$ the error is $O(q^2) = O(q^{(d+1)/2})$, much smaller than the $q^{d+1/2}$ allowed by Chebotarev. The reason is that W is a quadric, whose only nontrivial cohomology sits in the middle degree. Theorem 6.2 is equivalent to Kuz'min's formula for irreducible sextics with the coefficients of t^5 and t^4 prescribed [13, 15]. The derivation via W is included because it is short and because the code checks it independently.

7 Numerical verification

All computations use prime fields $q = p$. They were done with the code in the accompanying repository, which factors every element of $\mathcal{I}_f$ by distinct-degree factorization. The implementation was cross-checked against trial division.

7.1 Exact counts

Table 2 lists a sample of exact counts; the full table has one row for each (d, p, f) with $d \leq 6$. For $d \leq 3$, every computed value agrees with Theorem D. For $d = 4, 5, 6$ the normalized error $(2dN_{\text{irr}} - q^{d+1})/q^d$ stays bounded, consistent with Theorem C (Figure 1).

7.2 The twisted identity

For $d \leq 4$ and small p we enumerated all of $\mathbb{F}_{p^{2d}}$ and counted $\#W(\mathbb{F}_q)$ and $E(f)$ directly. The identity of Theorem B held in every case. $|\#W(\mathbb{F}_q) - q^{d+1}|$ was always far below the Cafure–Matera bound (`data/twisted_identity.csv`).

7.3 Factorization statistics

By Theorem A and Chebotarev, the proportion of $\mathbf{a}$ for which $F(t, \mathbf{a})$ has factorization type λ tends to $P(\lambda) = 1/z_\lambda$. Figure 2 compares the observed frequencies with $P(\lambda)$ for $(d, p) = (2, 47), (3, 31), (4, 19), (5, 13), (6, 13)$.

d	q	f	N_{irr}	$q^{d+1}/2d$	$\frac{2dN - q^{d+1}}{q^d}$	formula
1	3	t	3	4.5	-1.000	✓
1	43	t	903	924.5	-1.000	✓
1	43	$t + 34$	903	924.5	-1.000	✓
1	97	t	4656	4704.5	-1.000	✓
1	97	$t + 93$	4656	4704.5	-1.000	✓
2	5	t^2	30	31.2	-0.200	✓
2	5	$t^2 + 3t + 1$	30	31.2	-0.200	✓
2	23	t^2	3036	3041.8	-0.043	✓
2	23	$t^2 + 8t + 7$	3036	3041.8	-0.043	✓
2	47	t^2	25944	25955.8	-0.021	✓
2	47	$t^2 + 6t + 12$	25944	25955.8	-0.021	✓
3	7	t^3	398	400.2	-0.038	✓
3	7	$t^3 + 5t^2 + 4t + 6$	407	400.2	+0.120	✓
3	19	t^3	21714	21720.2	-0.005	✓
3	19	$t^3 + 11t^2 + 6t + 3$	21657	21720.2	-0.055	✓
3	31	t^3	153910	153920.2	-0.002	✓
3	31	$t^3 + 7t^2 + 19t + 24$	154075	153920.2	+0.031	✓
4	11	t^4	20420	20131.4	+0.158	
4	11	$t^4 + 5t^2 + 7t + 5$	20155	20131.4	+0.013	
4	17	t^4	176424	177482.1	-0.101	
4	17	$t^4 + 5t^3 + 15t^2 + 2t + 4$	177375	177482.1	-0.010	
4	19	t^4	310410	309512.4	+0.055	
4	19	$t^4 + 18t^3 + 11t^2 + 6t + 12$	309669	309512.4	+0.010	
5	11	t^5	175854	177156.1	-0.081	
5	11	$t^5 + 8t^4 + 4t^3 + 8t^2 + t + 2$	176995	177156.1	-0.010	
5	13	t^5	482100	482680.9	-0.016	
5	13	$t^5 + t^4 + 7t^3 + 4t^2 + 9t + 12$	482642	482680.9	-0.001	
6	13	t^6	5236204	5229043.1	+0.018	
6	13	$t^6 + 7t^5 + 5t^4 + 9t^3 + 5t^2 + 12$	5228155	5229043.1	-0.002	

Table 2: Exact values of $N_{\text{irr}}(f)$ for $f = t^d$ and a random f . “Formula” is Theorem D where it applies. Source: `data/interval_counts.csv`.

7.4 Exhaustive search over f

For small p we ran over every class of f , using translation $t \mapsto t + b$ to set $c_{d-1} = 0$ when $p \nmid d$. Table 3 gives the minimum of N_{irr} . It is positive in every case, including small characteristics $p \leq 2d$, which are outside Theorems A and C.

8 Discussion

Large q versus large d . Theorems A–D concern fixed d and growing q . This is the regime where the Riemann hypothesis for varieties over finite fields gives square-root cancellation. The closer analogue of the classical conjecture fixes q and lets $d \rightarrow \infty$. That requires an irreducible polynomial of degree $2d$ with its top $d - 1$ coefficients prescribed. Known results on prescribed coefficients [17, 10] allow roughly $n/4$ prescribed coefficients in degree n for large q , and a small positive proportion for every q . Half of the coefficients is out of reach of current methods, so we make no claim there.

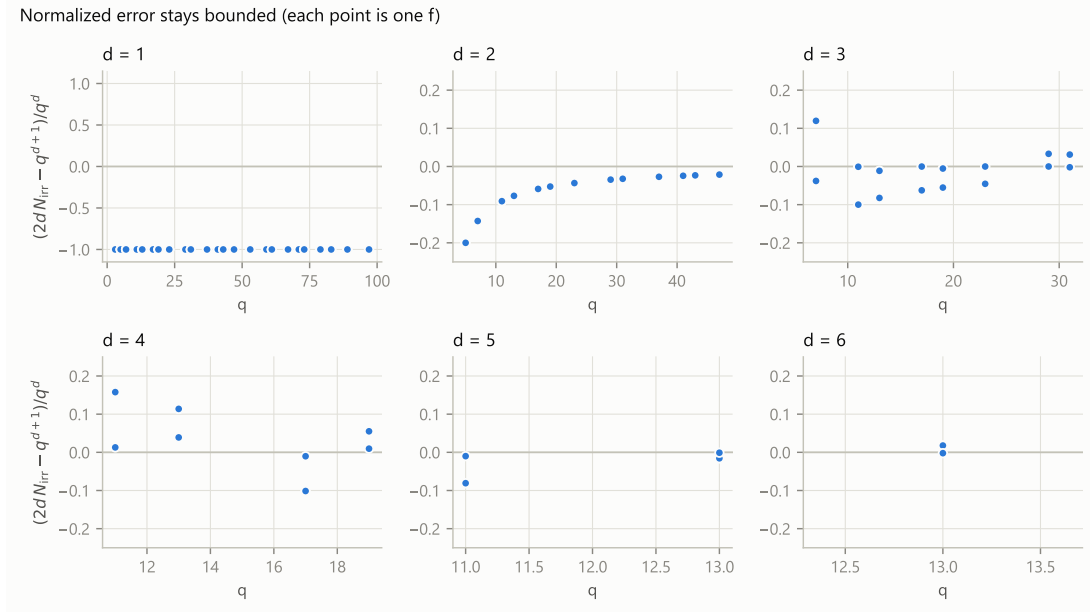


Figure 1: Normalized error $(2dN_{\text{irr}} - q^{d+1})/q^d$ against q , for $p > 2d$. The values stay bounded, as Theorem C requires. For $d = 1$ the error is exactly $-q$. For $d = 2, 3$ the values tend to 0, in line with the $O(q^{(d+1)/2})$ error of Theorem D.

Small characteristic. For $p \leq 2d$ the proofs of Propositions 3.2 and 3.3 do not apply as written. By [2] the asymptotic still holds for odd p . Theorem D(ii) covers $p = 3, d = 2$. The data of Table 3 show no failure.

Open problems.

- (1) Prove $N_{\text{irr}}(f) \geq 1$ for all q when $d \geq 4$, closing the gap between the computations (small q) and Corollary 5.3 (large q).
- (2) Find an explicit threshold of the form $q > C^{d \log d}$ or better, using Betti-number bounds for W .
- (3) Shorter intervals: find the least e such that $\{f^2 + s : \deg s \leq e\}$ contains an irreducible for all large q . By [2] the answer is 1 or 2 for large q , so the real question is uniformity in q and d .
- (4) Higher powers: the interval $\{f^k + s : \deg s \leq (k-1)d\}$ for $k \geq 3$.

A Errata to version v5

- E1. **Novelty.** The main theorem of v5, $N_{\text{irr}} = q^{d+1}/(2d) + O_d(q^{d+1/2})$, is the case $k = 2d, m = d$ of [2, Thm. 2.3]. That theorem also holds without the squarefree hypothesis and for every odd p . v5 cited [2] only as a possible source of sharper bounds.
- E2. **Primitivity.** v5 used the equivalence “imprimitive $\Leftrightarrow F = g \circ h$ ” for the Galois group of F over $\overline{\mathbb{F}}_q(\mathbf{a})$. That equivalence holds for $G(t) - x$ with x transcendental and appearing only in the constant term, not for a general polynomial over a field. The final dimension count also treated the coefficients of g , which lie in an algebraic extension of $\overline{\mathbb{F}}_q(\mathbf{a})$, as free parameters in $\mathbb{A}^n$. Fixed in Lemma 2.3 and Proposition 3.2.
- E3. **Transposition.** v5 claimed that near $\mathbf{a} = 0$ the discriminant has branches $\{P(\alpha_j) = 0\}$ along which two roots coalesce. This ignores the linear term $P'(\alpha_j)u$ and the parameter a_d .

Factorization types in $\mathcal{I}_{t^d}$ follow the cycle types of S_{2d}

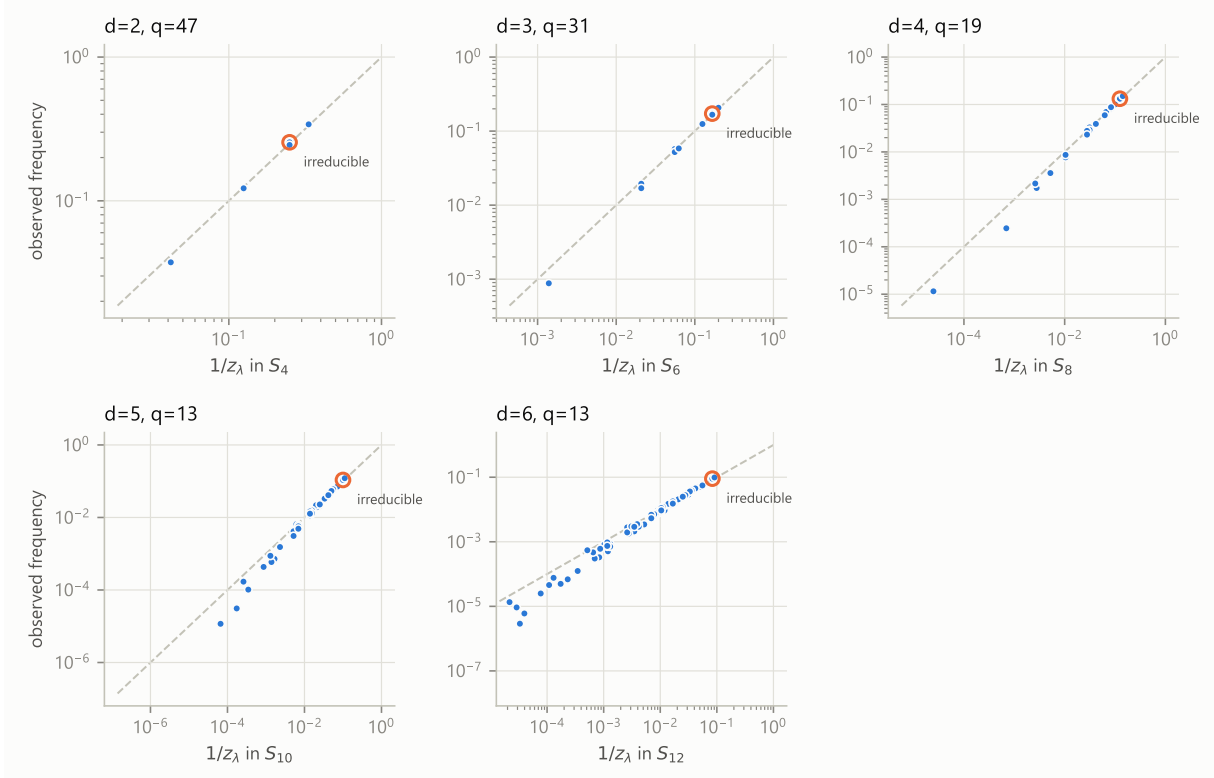


Figure 2: Observed frequency of each factorization type in $\mathcal{I}_{t^d}$ against the S_{2d} cycle-type probability $1/z_\lambda$. The ringed point is the irreducible type. Points lie near the diagonal, as full S_{2d} monodromy predicts. The rarest types, with many small factors, fall below it at these small q ; this is the $O(q^{-1/2})$ correction.

When $P(\alpha_j) = 0$ but $P'(\alpha_j) \neq 0$, the double root splits. The appeal to Picard–Lefschetz was not justified in characteristic p . The remark that $p > 2d$ is needed for $f'(\alpha_j) \neq 0$ is also wrong, since squarefree suffices. Fixed in Proposition 3.3, which does not need f squarefree.

- E4. **Effective Chebotarev.** The bound was stated with the permutation sheaf $\pi_* \overline{\mathbb{Q}}_\ell$. Traces of Frobenius on that sheaf count fixed points, so they detect linear factors, not $2d$ -cycles. Counting $2d$ -cycles needs all irreducible characters of S_{2d} that are nonzero on the class, which are the $2d$ hook characters. The cited “[11], Thm. 9.2.6” is not an effective Chebotarev theorem. Replaced by Theorem B.
- E5. **Betti-number bound.** v5 attributed $B(\mathcal{F}) \leq N(2\delta + 1)^n$ to [12, Thm. 12]. That theorem bounds Betti numbers of Artin–Schreier and Kummer sheaves $\mathcal{L}_\psi(f) \otimes \mathcal{L}_\rho(G)$, not of pushforwards of finite covers. The general bounds for compatible systems in [12] are not explicit. So the threshold $(8d)^{2d+6}$ was unproved. Replaced by Corollary 5.3.
- E6. **Error term bookkeeping.** In Step 4 of v5 the unspecified $O_{N,n}(q^{n-1})$ term of the Chebotarev bound was replaced by a Lang–Weil count of $\mathcal{D}(\mathbb{F}_q)$. These are different quantities, and the latter is not needed because $N_{\text{irr}} = N_{\text{irr}}^\circ$.
- E7. **GOS formula.** Equation (4.1) of v5 wrote $\chi(\mathbb{P}^1, j_! \mathcal{F}) = 2r - \sum(\text{drop}_v + \text{Sw}_v)$. That is the formula for j_* , not $j_!$. The section was not used in the proof and is omitted.
- E8. **Citations.** Bary-Soroker’s paper is *Proc. AMS* 137 (2009), 73–83, not vol. 140 (2012),

d	q	$p > 2d$	classes of f	$\min N_{\text{irr}}$	$\max N_{\text{irr}}$	$q^{d+1}/2d$
2	3	no	1	6	6	6.8
2	5	yes	1	30	30	31.2
2	7	yes	1	84	84	85.8
2	11	yes	1	330	330	332.8
2	13	yes	1	546	546	549.2
3	3	no	9	11	15	13.5
3	5	no	5	99	107	104.2
3	7	yes	7	391	407	400.2
3	11	yes	11	2418	2458	2440.2
3	13	yes	13	4730	4786	4760.2
4	3	no	9	27	37	30.4
4	5	no	25	375	402	390.6
4	7	no	49	2058	2131	2100.9
4	11	yes	121	20035	20420	20131.4
4	13	yes	169	46212	46818	46411.6
5	3	no	27	63	79	72.9
5	5	no	625	1450	1650	1562.5
5	7	no	343	11654	11893	11764.9
6	3	no	243	156	204	182.2
6	5	no	625	6383	6629	6510.4

Table 3: Minimum and maximum of $N_{\text{irr}}(f)$ over all f , for small p . Source: `data/exhaustive_min_counts.csv`.

and it works over PAC fields rather than via Chebotarev over finite fields. Several bibliography entries were never cited. The Arnold–Gusein-Zade–Varchenko reference was dated inconsistently.

- E9. **Numerical evidence.** v5 inferred that $N_{\text{irr}} \geq 1$ “for all q and d ” from three small examples. Section 7 replaces them with systematic data. The general claim remains open for $d \geq 4$.
- E10. **Version-history remarks** in the body of v5 (“erroneously used in earlier versions”) have been moved to this appendix.

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